

SeismoFNO: Physics-Grounded Neural Operators for Seismic Structural Response

A research brief on methodology, experiments, and open problems

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Why this problem

Simulating a multi-story building's response to an earthquake—tracking floor displacements across a 20-second strong shaking event—takes OpenSeesPy between 30 and 200 ms per structure depending on nonlinear plastic yielding. Regional seismic risk assessment involves tens of thousands of unique buildings across an urban grid; running exhaustive nonlinear time-history analyses for every inventory structure is computationally prohibitive. Neural operators offer an alternative: learn the mapping from ground motion input to multi-story dynamic response once, then evaluate near-instantly.

The governing elastoplastic equation is a coupled second-order matrix differential equation:

$$\mathbf{M} \ddot{\mathbf{u}}(t) + \mathbf{C} \dot{\mathbf{u}}(t) + \mathbf{f}_{\text{int}}(\mathbf{u}(t), \dot{\mathbf{u}}(t)) = -\mathbf{M} \mathbf{1} a_g(t) \quad (1)$$

From an operator-learning standpoint, this is a mapping between infinite-dimensional function spaces conditioned on structural parameters. However, civil structures violate standard FNO assumptions: variable story count, non-uniform stiffness, and hysteretic stiffness degradation that shifts fundamental natural frequencies during shaking.

1. Foundations: rigorous ground-truth validation (EXP1–EXP3)

Before training any surrogate, EXP1 validated single-degree-of-freedom elastoplastic dynamics and Bouc-Wen hysteretic loops against analytical solutions. EXP2 verified continuous hysteretic energy dissipation tracking. EXP3 established a zero-leakage data partition splitting simultaneously by structural archetype and ground motion identity:

Partition	Structural Archetypes	Earthquake Records	n	Scientific Evaluation Target
In-distribution (ID)	5 known archetypes ($T_1 \leq 0.85$ s)	RSN0001–08	120	Ordinary within-envelope accuracy
OOD-A	5 known archetypes ($T_1 \leq 0.85$ s)	RSN0011–12 (held-out)	300	Unseen ground motion generalization
OOD-B	5S_T120 only (held-out, $T_1 = 1.20$ s)	RSN0001–08	240	Unseen structural flexibility (extrapolation)
OOD-C	5S_T120 (held-out, $T_1 = 1.20$ s)	RSN0011–12 (held-out)	60	Combined structural + seismic extrapolation

2. The zero-padding failure mode (EXP4)

A standard 2D Fourier Neural Operator (Li et al., 2021) was configured on a fixed 5×1024 tensor, zero-padding shorter buildings to 5 stories. On 5-story buildings, it reached 19.29% median Relative L_2 error. On 3-story buildings, it collapsed to **99.60% median error**.

The failure is representational: discrete 2D spatial Fourier transforms assume periodic boundary conditions across the lattice. Zero-padding creates a sharp step discontinuity at the boundary between real and phantom floors. Global spectral kernels spread this discontinuity as Gibbs ringing back down into the physical floors. No duration of training resolves this because the representation violates physical geometry.

3. Graph topology eliminates the artifact (EXP5)

Replacing the Cartesian grid with a topology-native graph—where nodes are floor slabs and edges are physical columns carrying stiffness—dropped 3-story median error from 99.60% to **22.09%**. Each operator block combines spatial graph message-passing with a 1D temporal Fourier convolution. Parameters dropped from 4,735,187 to 674,115.

However, an out-of-distribution failure emerged on the flexible archetype 5S_T120 ($T_1 = 1.20$ s; training upper limit $T_1 = 0.85$ s): EXP5 produced **35.21% peak displacement error** and **115.70% Relative L_2 error**. Forensic FFT analysis revealed the unconditioned model oscillated at ~ 1.12 Hz instead of the true 0.83 Hz. Over a 20-second record, a 0.29 Hz discrepancy drifts the predicted and true signals into phase opposition, generating theoretical relative L_2 errors near $\sqrt{2} \approx 141\%$.

4. Modal conditioning via FiLM (EXP6)

Prior to shaking, an engineer can compute undamped modal invariants ($T_1\text{--}T_3$, $\omega_1\text{--}\omega_3$) directly from the pre-earthquake stiffness and mass matrices $[\mathbf{K}, \mathbf{M}]$. Injecting these invariants via Feature-wise Linear Modulation (FiLM):

$$[\gamma, \beta] = \text{MLP}(\mathbf{c}), \quad \mathbf{h}_{\text{mod}} = (1 + \gamma) \odot \mathbf{h} + \beta \tag{2}$$

Modulating both the spatial message-passing and temporal Fourier channels produced a decisive reduction in peak structural displacement error on unseen structures:

Architecture	Parameters	ID Peak	OOD-A Peak	OOD-B Peak	OOD-C Peak
EXP4 FNO2D (zero-padded)	4,735,187	—	51.74%	3.17%†	50.72%
EXP5 Baseline GNO (unconditioned)	674,115	12.70%	15.86%	35.21%	38.66%
EXP6-B T_1 -conditioned GNO	725,059	2.09%	8.81%	13.47%	14.36%
EXP6-C Multi-modal GNO ($T_1\text{--}3$, $\omega_1\text{--}3$)	727,619	8.87%	7.63%	13.06%	17.01%
EXP6-D Shuffled T_1 (falsification ablation)	725,059	2.65%	8.94%	24.33%	36.16%

†EXP4 OOD-B value is anomalous due to structural overlap in that initial split, corrected in EXP5/EXP6.

Falsification ablation: Permuting conditioning vectors across batch samples (EXP6-D) caused OOD-B peak error to jump from 13.47% back to **24.33%**, and OOD-C from 14.36% to **36.16%**. This demonstrates the model genuinely extracts physical modal correspondence rather than benefiting from auxiliary parameter capacity.

5. Architectural limitations: the phase drift problem

Relative L_2 error on OOD-B remains above 100% despite modal conditioning. Peak amplitude is controlled, but waveform phase is not. Global 1D Fourier layers apply a static complex weight matrix across the entire 20.48-second record—they cannot warp their frequency basis dynamically as the structure softens during plastic yielding. This is an architectural limitation of global Fourier representations, pointing to causal state-space models (S4, Mamba) or neural ODEs as the necessary next step.

6. Inference latency & throughput benchmarks

Engine / Configuration	Batch	Latency	Throughput	Speedup vs. OpenSeesPy
OpenSeesPy 5-story NLTHA (reference)	1	54.68 ms	18.3 sim/s	1.00×
EXP4 FNO2D (fixed-grid)	1	6.60 ms	151.5 sim/s	8.28×
EXP5 Spatiotemporal GNO	1	16.25 ms	61.5 sim/s	3.36×
EXP6-B T_1 -conditioned GNO (single)	1	21.45 ms	46.6 sim/s	2.55×
EXP6-B T_1 -conditioned GNO (batched)	32	30.19 ms	1,060 sim/s	~58× (throughput)

Measured on Apple Silicon GPU (MPS) with process-wide device synchronization. Latencies are medians over 50 runs.

7. Reproducibility & scientific integrity

The project enforces zero-leakage data split verification, 305 passing unit tests (pytest), frozen model weights with SHA-256 integrity checks, and an automated forensic audit script. All numbers are computed live from evaluation CSVs at `results/experiments/exp6/evaluation/` and recorded in `results/canonical/canonical_results.json`.